CS422- Spring 26 - Project - Individual

Due: June 13th, end of the day - Chicago Time

1 Project Notebook

Please provide the following regarding the project notebook (Note: Python 3.11+/scikit-learn 1.5+/ONNX 1.16+/onnxruntime 1.20+):

- **A Scikit-Learn based Pipeline for training with the provided data in csv format - model research/fit development.**

  Why this is needed: Scikit-Learn Pipelines allow you to chain preprocessing steps (scaling, imputation, feature selection) and the model together. This ensures:

  - No data leakage between training and testing
  - Easy experimentation with different preprocessing combinations
  - Reproducible workflows that can be shared and validated
  - Built-in cross-validation with GridSearchCV for hyperparameter tuning

- **An ONNX based Model for testing with a runtime session in ONNX format - model deployment/edge inference.**

  Why this is needed: ONNX (Open Neural Network Exchange) converts your trained scikit-learn model into a format that can run anywhere, not just in Python:

  - **Faster inference** - Optimized execution (2-5x faster than scikit-learn)
  - **Cross-platform** - Runs in C++, Java, C#, JavaScript, and on mobile devices
  - **Edge deployment** - Works on Raspberry Pi, Android, iOS without a Python environment
  - **Smaller footprint** - Reduced memory usage for production environments
  - **Hardware acceleration** - Can leverage GPU, AVX, and NEON instructions

  Note: You will train with scikit-learn (research phase), then convert to ONNX and test inference with ONNX Runtime (deployment phase). Both must work identically.

2 Project Report

Please provide the following regarding the project report (Note: Can be inline with the Project Notebook):

- Abstract - Research summary, findings, and next steps.
- Overview - Problem statement, relevant literature, proposed methodology.
- Data Processing - Pipeline details, data issues, assumptions/adjustments.
- Data Analysis - Summary statistics, visualization, feature extraction.
- Model Training - Feature engineering, evaluation metrics, model selection.
- Model Validation - Testing results, performance criteria, biases/risks.
- Conclusion - Positive/Negative results, recommendations, caveats/cautions.
- Data Sources - Links, downloads, access information.
- Source Code - Listings, documentation, dependencies (open-source).
- Bibliography - Reference citations (Chicago style - AMS/AIP or ACM/IEEE).